

Study on Covariance Matrix Estimation

Part 1: Problem Statement and Model Review

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The Estimation Problem in High Dimension

1 The True Covariance and the Natural Estimator

We observe n daily returns of p assets. Let $r_t = (r_{1t}, \dots, r_{pt})^\top$ be the return vector at date t , assumed to be centered.

The true covariance matrix Σ is a $p \times p$ symmetric positive definite matrix. It contains all information about risks and dependencies between assets:

$$\Sigma = \mathbb{E}[r_{it} r_{jt}]$$

where Σ_{ii} is the variance of asset i and Σ_{ij} is the covariance between assets i and j .

Σ contains p variances on the diagonal and $\frac{p(p-1)}{2}$ off-diagonal covariances, for a total of $\frac{p(p+1)}{2}$ distinct parameters to estimate.

The true matrix Σ is unknown and unobservable. We only have access to the n observed returns.

The natural estimator is the sample covariance, computed pairwise:

$$\hat{S}_{ij} = \frac{1}{n} \sum_{t=1}^n r_{it} r_{jt} \quad \implies \quad S = \frac{1}{n} X^\top X$$

where X is the $n \times p$ matrix of observed returns. Each entry $\hat{S}_{ij}$ is computed by an exact closed-form formula: it is the average of the observed cross-products.

2 Why Each Pairwise Estimate Is Imperfect: Individual Sampling Uncertainty

We observe a finite sample of n realizations of a random process. This sample is one among infinitely many possible realizations. If the experiment were repeated on n other days, we would obtain a slightly different $\hat{S}_{ij}$. This is sampling uncertainty — it is unavoidable with a finite sample.

The law of large numbers guarantees convergence:

$$\hat{S}_{ij} \xrightarrow{n \rightarrow \infty} \Sigma_{ij}.$$

To quantify this uncertainty, we compute the variance of the estimator. Using variance properties,

$$\text{Var}(\hat{S}_{ij}) = \text{Var}\left(\frac{1}{n} \sum_{t=1}^n r_{it} r_{jt}\right) = \frac{1}{n^2} \text{Var}\left(\sum_{t=1}^n r_{it} r_{jt}\right).$$

Assuming observations are independent and identically distributed, the variance of a sum is the sum of the variances:

$$\text{Var}\left(\sum_{t=1}^n r_{it} r_{jt}\right) = \sum_{t=1}^n \text{Var}(r_{it} r_{jt}) = n \text{Var}(r_{it} r_{jt}).$$

This gives:

$$\text{Var}(\hat{S}_{ij}) = \frac{1}{n} \text{Var}(r_{it} r_{jt}).$$

We expand $\text{Var}(r_{it} r_{jt})$ using $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$:

$$\mathbb{E}[\Sigma_{ij}^2] = \mathbb{E}[(r_{it} r_{jt})^2] - (\mathbb{E}[r_{it} r_{jt}])^2.$$

By Wick's theorem (which expresses the fourth-order moments of a Gaussian vector in terms of its second-order moments):

$$\mathbb{E}[(r_{it} r_{jt})^2] = \Sigma_{ii} \Sigma_{jj} + 2 \Sigma_{ij}^2 - \Sigma_{ij}^2 = \Sigma_{ii} \Sigma_{jj} + \Sigma_{ij}^2.$$

Therefore:

$$\boxed{\text{Var}(\hat{S}_{ij}) = \frac{\Sigma_{ii} \Sigma_{jj} + \Sigma_{ij}^2}{n}}.$$

This is the residual uncertainty on the estimation of the covariance between i and j over a sample of size n . Estimation uncertainty decreases with n .

Two special cases:

- Diagonal term ($i = j$):

$$\text{Var}(\hat{S}_{ii}) = \frac{2 \Sigma_{ii}^2}{n}.$$

- Uncorrelated assets ($\Sigma_{ij} = 0$):

$$\text{Var}(\hat{S}_{ij}) = \frac{\Sigma_{ii} \Sigma_{jj}}{n}.$$

Estimation uncertainty decreases with n .

3 The Total Error on S : Quadratic Accumulation in p

There are $\frac{p(p+1)}{2}$ parameters to estimate simultaneously, each with its own uncertainty. The total error on S is the sum of all these uncertainties:

$$\mathcal{E}_{\text{tot}} = \mathbb{E}[\|S - \Sigma\|_F^2] = \sum_{i,j} \text{Var}(\hat{S}_{ij}).$$

We separate the two groups of parameters. Let σ^2 denote the typical variance common to all assets.

1. The p diagonal terms:

$$\mathcal{E}_{\text{diag}} = \sum_{i=1}^p \text{Var}(\hat{S}_{ii}) = p \cdot \frac{2\sigma^4}{n} = \frac{2p\sigma^4}{n}.$$

There are p diagonal terms, each with variance $\frac{2\sigma^4}{n}$.

2. The $\frac{p(p-1)}{2}$ off-diagonal terms. With ρ the average correlation between assets:

$$\text{Var}(\hat{S}_{ij}) = \frac{\sigma^4 + \rho^2 \sigma^4}{n} = \frac{\sigma^4(1 + \rho^2)}{n} \quad (i \neq j),$$

$$\mathcal{E}_{\text{off-diag}} = \frac{p(p-1)}{2} \cdot \frac{\sigma^4(1 + \rho^2)}{n}.$$

Assembly:

$$\mathcal{E}_{\text{tot}} = \frac{2p\sigma^4}{n} + \frac{p(p-1)}{2} \cdot \frac{\sigma^4(1 + \rho^2)}{n} = \frac{\sigma^4}{n} \left[2p + \frac{p(p-1)(1 + \rho^2)}{2} \right].$$

For large p , the term $2p$ is negligible compared to $\frac{p(p-1)}{2} \approx \frac{p^2}{2}$. The dominant behavior is:

$$\mathcal{E}_{\text{tot}} \approx \frac{p^2 \sigma^4(1 + \rho^2)}{2n} = O\left(\frac{p^2}{n}\right).$$

The total error is quadratic in p and linear in $\frac{1}{n}$. If p doubles, the error quadruples.

4 Eigenvalues: What They Are and Why They Matter

So far we have discussed the error on the entries of S . But in finance, S is used in global matrix operations (Markowitz optimization, minimum variance computation, etc.). All these operations go through the spectral decomposition of S , i.e., its eigenvectors and eigenvalues.

An eigenvector u_k and its eigenvalue λ_k satisfy:

$$\Sigma u_k = \lambda_k u_k.$$

Applying Σ to u_k amounts to multiplying by the scalar λ_k : the vector does not change direction, only scale.

The eigenvalue λ_k has a direct financial interpretation: it is the **variance of the portfolio** u_k . If u_k is normalized ($\|u_k\|^2 = 1$):

$$\text{Var}(u_k^\top r) = u_k^\top \Sigma u_k = u_k^\top (\lambda_k u_k) = \lambda_k \|u_k\|^2 = \lambda_k.$$

The larger λ_k , the riskier this portfolio.

The spectral decomposition raises the following question: among all possible portfolios, are there special directions where risk is “pure”, i.e., where Σ acts simply as multiplication by a scalar? Yes — these are the eigenvectors.

The spectral theorem guarantees that Σ can be fully decomposed:

$$\Sigma = U \Lambda U^\top = \sum_{k=1}^p \lambda_k u_k u_k^\top$$

where $U = [u_1, \dots, u_p]$ is the orthogonal matrix of eigenvectors and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$.

Each term $\lambda_k u_k u_k^\top$ is the contribution of the k -th risk factor. Σ is the superposition of p independent pure risks.

Its inverse, at the core of Markowitz:

$$\Sigma^{-1} = U \Lambda^{-1} U^\top = \sum_{k=1}^p \frac{1}{\lambda_k} u_k u_k^\top.$$

Inversion replaces each λ_k by $\frac{1}{\lambda_k}$. If λ_k is underestimated, $\frac{1}{\lambda_k}$ becomes very large: the apparently low-risk direction receives an enormous and erroneous weight in the optimization.

5 How Entry-Level Uncertainty Propagates to the Spectrum

Eigenvalues are nonlinear functions of all entries simultaneously. Computing an eigenvalue amounts to solving $\det(S - \lambda I) = 0$, a degree- p polynomial that mixes all $\frac{p(p+1)}{2}$ entries. Each entry imports its uncertainty into this mixture.

Let $E = S - \Sigma$ be the error matrix. Each entry $E_{ij} = \hat{S}_{ij} - \Sigma_{ij}$ is the sampling uncertainty on pair (i, j) : it is centered and small.

Weyl’s theorem bounds the displacement of eigenvalues:

$$|\ell_k(S) - \lambda_k(\Sigma)| \leq \|E\|_{\text{op}} \quad \forall k,$$

where $\|E\|_{\text{op}} = \lambda_{\max}(E)$ is the operator norm of E . It measures the size of E as a linear operator and controls the displacement of all eigenvalues.

The operator norm is written as a maximization over all unit vectors:

$$\|E\|_{\text{op}} = \max_{\|u\|=1} u^\top E u = \max_{\|u\|=1} \sum_{i=1}^p \sum_{j=1}^p u_i E_{ij} u_j.$$

For a fixed vector u , the noise $u^\top E u$ is a sum of p^2 terms $u_i E_{ij} u_j$, each small.

But by maximizing over all unit vectors in a p -dimensional space, we find the direction where the p^2 uncertainties accumulate most favorably:

$$\|E\|_{\text{op}} = \max_{\|u\|=1} u^\top E u \approx \frac{\sigma^2}{\sqrt{n}} \cdot \sqrt{p} = \sigma^2 \sqrt{\frac{p}{n}} = \sigma^2 \sqrt{c}.$$

Each E_{ij} has standard deviation $\approx \frac{\sigma^2}{\sqrt{n}}$. The maximum over the unit sphere of dimension p accumulates these terms and produces a result of order $\sigma^2\sqrt{c}$, which does not tend to zero if $c > 0$.

Each entry of E is small with standard deviation $\frac{\sigma^2}{\sqrt{n}}$ that tends to zero. Yet the operator norm is of order $\sigma^2\sqrt{c}$, which does not tend to zero if $c > 0$. The difference is a factor $\sqrt{p}$ coming from the dimension of the maximization space. **The more assets we have, the larger the unit sphere, and the more likely it is to find a direction that concentrates the noise.**

6 The Marčenko–Pastur Law: Exact Quantification of Spectral Bias

To quantify this bias precisely, Marčenko and Pastur (1967) studied the reference case $\Sigma = I$ (identity matrix) — no real structure, all assets independent with unit variance. In this case the true answer is clear: all eigenvalues of Σ equal 1. Any deviation of the eigenvalues of S from 1 is therefore pure sampling noise.

When $n, p \rightarrow \infty$ with $\frac{p}{n} \rightarrow c$, the distribution of eigenvalues of S converges to the **Marčenko–Pastur law**:

$$f_{\text{MP}}(\lambda; c) = \frac{1}{2\pi c\lambda} \sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}, \quad \lambda \in [\lambda_-, \lambda_+],$$

with bounds $\lambda_{\pm} = (1 \pm \sqrt{c})^2$.

If $c > 1$, an atom of mass $1 - \frac{1}{c}$ at $\lambda = 0$ corresponds to the null eigenvalues of the singular matrix.

By the Bai–Yin theorems:

$$\ell_{\max}(S) \xrightarrow{p, n \rightarrow \infty} \lambda_{\max}(\Sigma) \cdot (1 + \sqrt{c})^2 \quad \text{systematic overestimation,}$$

$$\ell_{\min}(S) \xrightarrow{p, n \rightarrow \infty} \lambda_{\min}(\Sigma) \cdot (1 - \sqrt{c})^2 \quad \text{systematic underestimation.}$$

All intermediate eigenvalues are distorted according to f_{MP} .

- Large eigenvalue multiplied by $(1 + \sqrt{c})^2 > 1$: systematic overestimation.
- Small eigenvalue multiplied by $(1 - \sqrt{c})^2 < 1$: systematic underestimation.

The small eigenvalue, poorly estimated downward, gives an enormous $\frac{1}{\lambda_{\min}}$ in S^{-1} : the apparently low-risk direction takes an overwhelming and erroneous weight in Markowitz optimization.

Linear Shrinkage: Ledoit & Wolf (2004)

1 Starting Point

We know that the eigenvalues of S are distorted: large ones are overestimated and small ones are underestimated.

The idea of Ledoit and Wolf in 2004 is to find a single scalar constant that summarizes the global bias and allows the entire spectrum to be corrected in one operation.

2 Linear Shrinkage: Pulling S Toward a Target

Shrinkage consists of replacing S by a convex combination between S and a target T , the latter being a matrix deliberately chosen to be simple and well-conditioned.

Shrunk estimator:

$$\hat{\Sigma} = \rho T + (1 - \rho)S, \quad \rho \in [0, 1].$$

ρ is the shrinkage intensity, the central parameter of the model. The convex combination pulls S toward T with strength ρ .

Effect on the spectrum: if S has eigenvalues that are too dispersed and T is a matrix with all equal eigenvalues, then $\hat{\Sigma} = \rho T + (1 - \rho)S$ has eigenvalues less dispersed than those of S , but not completely flat. This is exactly the correction we are looking for.

3 The Spherical Target: Why This Choice?

The authors choose as target the **spherical matrix** $T = \bar{s} I_p$, where $\bar{s}$ is the average variance of the assets and I_p is the identity matrix:

$$\bar{s} = \frac{1}{p} \text{tr}(S) = \frac{1}{p} \sum_{i=1}^p \hat{S}_{ii} \quad \implies \quad T = \bar{s} I_p.$$

$\text{tr}(S)$ is the trace of S , the sum of its diagonal elements, i.e., the sum of the estimated variances of each asset. Dividing by p gives the average variance. The target is therefore a diagonal matrix where all assets have the same variance $\bar{s}$ and all covariances are zero.

Three reasons justify this choice:

1. **T is well-conditioned by construction.** $T = \bar{s} I_p$ has all its eigenvalues equal to $\bar{s} > 0$. Its condition number is 1, the best possible. It is always invertible, regardless of the dimension p .

2. **T requires only one estimated parameter.** T is entirely determined by $\bar{s}$, computed directly from the data. No additional assumptions, no parameter to calibrate.
3. **T is the target that minimizes the error under structural constraint.** Among all matrices of the form αI , $T = \bar{s} I_p$ is the one that minimizes $\|T - \Sigma\|_F^2$ in expectation. It is the best spherical approximation of Σ in the least-squares sense.

The target says: all assets have the same average variance and no correlation between them. This is a very strong and probably false assumption, but it introduces zero estimation noise. We trade a known bias for a massive reduction in noise, and shrinkage doses this trade-off via ρ .

4 The Optimization Problem: Finding the Optimal ρ

We now need to find the optimal value of ρ . The objective is to minimize the mean squared error between the shrunk estimator and the true covariance, in the Frobenius norm sense:

$$\rho^* = \arg \min_{\rho \in [0,1]} \mathbb{E} \left[\|\rho T + (1 - \rho)S - \Sigma\|_F^2 \right].$$

Let $\hat{\Sigma}(\rho) = \rho T + (1 - \rho)S$ and expand the squared norm:

$$\begin{aligned} \|\hat{\Sigma}(\rho) - \Sigma\|_F^2 &= \left\| \underbrace{\rho(T - S)}_{\text{bias introduced by shrinkage}} + \underbrace{(S - \Sigma)}_{\text{estimation error of } S} \right\|_F^2 \\ &= \rho^2 \|T - S\|_F^2 + 2\rho \langle T - S, S - \Sigma \rangle_F + \|S - \Sigma\|_F^2. \end{aligned}$$

Taking the expectation and differentiating with respect to ρ , then setting to zero:

$$\frac{\partial}{\partial \rho} \mathbb{E} [\|\hat{\Sigma}(\rho) - \Sigma\|_F^2] = 0 \quad \implies \quad 2\rho \mathbb{E} [\|T - S\|_F^2] + 2 \mathbb{E} [\langle T - S, S - \Sigma \rangle_F] = 0.$$

This gives the exact analytical solution:

$$\boxed{\rho^* = - \frac{\mathbb{E} [\langle T - S, S - \Sigma \rangle_F]}{\mathbb{E} [\|T - S\|_F^2]}.}$$

This is the ratio of two quantities:

- in the numerator, the covariance between the shrinkage error $(T - S)$ and the estimation error $(S - \Sigma)$
- in the denominator, the dispersion of S around T . Both quantities can be estimated directly from the data.

5 The Three Key Quantities: d^2 , b^2 and $\bar{s}$

The authors show that ρ^* can be rewritten as a function of three quantities estimable from the data.

Quantity 1: d^2 , the total dispersion of S around the target T

$$d^2 = \|S - T\|_F^2 = \sum_{i,j} (S_{ij} - T_{ij})^2.$$

Squared distance between S and T . Since T is diagonal with $\bar{s}$ on the diagonal and 0 off-diagonal:

$$d^2 = \underbrace{\sum_i (S_{ii} - \bar{s})^2}_{\text{dispersion of variances around their mean}} + \underbrace{\sum_{i \neq j} S_{ij}^2}_{\text{magnitude of covariances}}.$$

If S is already spherical ($d^2 \approx 0$), no shrinkage is needed.

Therefore d^2 measures how dispersed the eigenvalues of S are around their mean. This is exactly the spectral overdispersion described by Marčenko–Pastur.

A large d^2 means S is far from a spherical matrix — it contains either a lot of real structure, or a lot of noise.

Quantity 2: b^2 , the sampling variance of S

b^2 measures how much S fluctuates around its own expectation, i.e., the share of d^2 that is due to sampling noise rather than true structure.

It is the mean squared error between the individual rank-1 matrices $x_t x_t^\top$ and their mean S :

$$b^2 = \frac{1}{n^2} \sum_{t=1}^n \|x_t x_t^\top - S\|_F^2.$$

Each observation x_t contributes to S via the rank-1 product $x_t x_t^\top$. The mean of these n matrices is exactly $n \cdot S$. b^2 measures the variance of these contributions around their mean, i.e., the sampling variance of S in the matrix sense.

The authors show that b^2 admits a closed-form formula. Expanding $\|x_t x_t^\top - S\|_F^2$ and summing:

$$b^2 = \frac{1}{n^2} \left[\sum_{t=1}^n \|x_t\|^4 + n \|S\|_F^2 - 2 \sum_{t=1}^n x_t^\top S x_t \right].$$

Three terms:

- $\sum_t \|x_t\|^4 = \sum_t (x_t^\top x_t)^2$: sum of squared norms raised to the power two.
- $n \|S\|_F^2 = n \sum_{i,j} S_{ij}^2$: Frobenius norm of S scaled by n .
- $2 \sum_t x_t^\top S x_t$: quadratic form of each observation in S .

b^2 is the share of d^2 that is pure noise. If $b^2 \approx d^2$, all the dispersion of S around T is noise: strong shrinkage is needed ($\rho \approx 1$). If $b^2 \ll d^2$, the dispersion is mainly real signal: little shrinkage is needed ($\rho \approx 0$).

6 The Formula for Optimal ρ : The Central Result

$$\boxed{\rho^* = \frac{b^2}{d^2}}.$$

This is the central result of the paper. The ratio b^2/d^2 is the proportion of sampling noise in the total dispersion of S . It automatically lies in $[0, 1]$ since $b^2 \leq d^2$ by construction (noise cannot exceed total dispersion).

- $b^2/d^2 \approx 1$: strong shrinkage. All the dispersion of S is noise; S provides no true information beyond T .
- $b^2/d^2 \approx 0$: weak shrinkage. The dispersion of S is mainly real signal; S is already a good estimate.

7 The Final Estimator and Its Spectral Interpretation

The final shrunk estimator is:

$$\hat{\Sigma}_{\text{LW}} = \rho^* \bar{s} I_p + (1 - \rho^*) S.$$

It is symmetric positive definite by construction (convex combination of two SPD matrices). It is computable in $O(np^2)$, the same order as computing S .

At the spectral level, the eigenvalues of $\hat{\Sigma}_{\text{LW}}$ are:

$$\ell_k(\hat{\Sigma}_{\text{LW}}) = \rho^* \bar{s} + (1 - \rho^*) \ell_k(S).$$

Each eigenvalue of S is pulled toward the mean $\bar{s}$ with weight ρ^* . Large eigenvalues (overestimated) are reduced. Small ones (underestimated) are increased. The spectrum is compressed toward the center.

8 Properties of the Estimator

Ledoit & Wolf (2004) prove four important properties.

1. **Asymptotic consistency.** When $n \rightarrow \infty$ with p fixed: $\rho^* \rightarrow 0$ and $\hat{\Sigma}_{\text{LW}} \rightarrow S \rightarrow \Sigma$. In the classical regime, shrinkage disappears automatically.
2. **Oracle optimality.** $\hat{\Sigma}_{\text{LW}}$ is asymptotically equivalent to the oracle estimator that would know Σ and choose ρ^* accordingly. Nothing is lost by estimating ρ^* from the data.

3. **Guaranteed positive definiteness.** $\hat{\Sigma}_{\text{LW}} = \rho^* \bar{s} I + (1 - \rho^*) S$. If $\rho^* > 0$ and $\bar{s} > 0$, the matrix is strictly positive definite even if S is singular ($p \geq n$). $\hat{\Sigma}^{-1}$ always exists.
4. **Reduction of matrix MSE.** $\mathbb{E}[\|\hat{\Sigma}_{\text{LW}} - \Sigma\|_F^2] \leq \mathbb{E}[\|S - \Sigma\|_F^2]$ for all n, p, Σ . The shrunk estimator is always better than S in terms of matrix squared error.

9 Limitations of Linear Shrinkage

Linear shrinkage is the best affine correction (convex combination $\rho T + (1 - \rho)S$), but it is not the best possible correction. It rests on a strong assumption: the optimal correction is the same for all eigenvalues.

Fundamental problem: ρ^* is a single scalar. It pulls all eigenvalues of S toward $\bar{s}$ with the same strength ρ^* . But the bias of the eigenvalues of S is not uniform: large eigenvalues are overestimated by a factor $(1 + \sqrt{c})^2$ and small ones are underestimated by a factor $(1 - \sqrt{c})^2$. These two factors differ and depend on the position in the spectrum. A uniform shrinkage cannot optimally correct both biases simultaneously.

Linear shrinkage is therefore a compromise. The truly optimal correction would be a **nonlinear shrinkage**: applying a different function to each eigenvalue according to its position in the spectrum. This is what the following models do: ANLS (Ledoit–Wolf 2020) and QIS (Ledoit–Wolf 2022).

Analytical Nonlinear Shrinkage: Ledoit & Wolf (2020)

1 Fundamental Limitation of Linear Shrinkage

Linear shrinkage corrects the spectrum of S by pulling all eigenvalues toward the mean $\bar{s}$ with the same strength ρ^* .

Recall of spectral bias: by the Bai–Yin theorem, the eigenvalues of S are distorted according to their position in the spectrum:

$$\ell_k(S) \approx \lambda_k(\Sigma) \cdot \phi(\lambda_k, c)$$

The distortion function ϕ depends on the position λ_k in the spectrum and on the ratio $c = p/n$. It is not constant and varies from one eigenvalue to another: large eigenvalues are multiplied by $(1 + \sqrt{c})^2$ and small ones by $(1 - \sqrt{c})^2$. Intermediate eigenvalues are distorted by different factors according to Marčenko–Pastur.

Linear correction is therefore a compromise. It is not optimal because the bias depends on the position in the spectrum.

2 The Idea of Nonlinear Shrinkage: One Function per Eigenvalue

The ANLS estimator replaces the uniform shrinkage by an individual correction function applied to each eigenvalue of S separately.

We keep the eigenvectors of S : we trust them for directions. We only replace the eigenvalues:

$$\hat{\Sigma}_{\text{ANLS}} = \sum_{k=1}^p d_k u_k u_k^\top = U \text{diag}(d_1, \dots, d_p) U^\top$$

where $U = [u_1, \dots, u_p]$ are the eigenvectors of S and d_k is the corrected eigenvalue of the k -th direction.

The ANLS estimator finds the optimal d_k one eigenvalue at a time, exploiting the asymptotic structure of the Marčenko–Pastur law.

The problem becomes:

$$(d_1^*, \dots, d_p^*) = \arg \min_{d_1, \dots, d_p} \mathbb{E} \left[\left\| \sum_{k=1}^p d_k u_k u_k^\top - \Sigma \right\|_F^2 \right].$$

A problem of p parameters. Ledoit and Wolf show that each d_k^* can be expressed analytically

as a function of the spectrum of S and c , without estimating each parameter individually:

$$d_k^* = \frac{\ell_k}{(\pi c \ell_k f_m(\ell_k))^2 + (1 - c - \pi c \ell_k Hf_m(\ell_k))^2}.$$

It depends on two quantities for each eigenvalue ℓ_k :

- The local density $f_m(\ell_k)$ of eigenvalues around ℓ_k . If many eigenvalues are close to ℓ_k , the density is high, sampling noise is strong at that point, and more correction is needed.
- The Hilbert transform $Hf_m(\ell_k)$ of this density. It captures the local asymmetry of the spectrum around ℓ_k . If more eigenvalues lie to the left than to the right, Hf_m is negative and ℓ_k is pulled upward by its neighbors.

d_k^* is a corrected version of ℓ_k that accounts for its local spectral environment. ANLS adapts the correction to each eigenvalue based on the density and asymmetry of the spectrum at that point.

3 The Stieltjes Transform: The Central Tool of RMT

The Stieltjes transform is the fundamental tool that connects the distribution of eigenvalues of a large random matrix to its spectral properties.

For a probability distribution F on $\mathbb{R}$, its Stieltjes transform is:

$$m_F(z) = \int \frac{1}{\lambda - z} dF(\lambda), \quad z \in \mathbb{C}, \quad \text{Im}(z) > 0.$$

This is an integral in the complex plane. F is here the empirical distribution of eigenvalues of a large random matrix. Each eigenvalue ℓ_k has weight $1/p$. $m_F(z)$ encodes all spectral information of the matrix in an analytic function of one complex variable.

Marčenko and Pastur showed that the Stieltjes transform of the empirical eigenvalue distribution of S satisfies a functional equation linking $m_S(z)$ to $m_\Sigma(z)$, the Stieltjes transform of the true covariance.

The key point is the relationship between the Stieltjes transform and the eigenvalue density. Evaluating $m_F(z)$ in the limit as z approaches the real axis from above:

$$m_F(\lambda + i\varepsilon) \xrightarrow{\varepsilon \rightarrow 0^+} \underbrace{Hf(\lambda)}_{\text{real part: Hilbert transform of the density}} + i\pi \underbrace{f(\lambda)}_{\text{imaginary part: } \pi \text{ times the density}}.$$

This directly connects the Stieltjes transform to the local spectral density. These are precisely the two terms $f_m(\ell_k)$ and $Hf_m(\ell_k)$ in the formula for d_k^* : the imaginary and real parts of the Stieltjes transform of S evaluated at ℓ_k , which can be estimated from the observed eigenvalues of S .

4 The Formula for d_k^* : Intuitive Derivation

The authors show that in the asymptotic regime $p/n \rightarrow c$, the optimal oracle eigenvalue d_k^* , the one that minimizes the squared error for the k -th direction, is:

$$d_k^* = \frac{1}{m_\Sigma(\ell_k)}.$$

The optimal correction of ℓ_k is the inverse of the Stieltjes transform of Σ evaluated at ℓ_k .

Problem: m_Σ depends on Σ , which is unknown.

The Marčenko–Pastur equation links $m_S(z)$ to $m_\Sigma(z)$ via the Stieltjes transform:

$$m_S(\ell_k + i0^+) = Hf_m(\ell_k) + i\pi f_m(\ell_k)$$

where $f_m(\ell_k)$ is the empirical density of the eigenvalues of S in the neighborhood of ℓ_k , and $Hf_m(\ell_k)$ is the Hilbert transform of this density, capturing the local asymmetry of the spectrum around ℓ_k .

Substituting into the formula for d_k^* and taking the squared modulus:

$$d_k^* = \frac{\ell_k}{(\pi c \ell_k f_m(\ell_k))^2 + (1 - c - \pi c \ell_k Hf_m(\ell_k))^2}.$$

The ratio corrects ℓ_k by accounting for the local density and asymmetry of the spectrum around that eigenvalue.

It depends on two quantities for each eigenvalue ℓ_k :

- The local density $f_m(\ell_k)$ of eigenvalues around ℓ_k (gives the amplitude). If many eigenvalues are close to ℓ_k , the density is high, sampling noise is strong at that point, and more correction is needed.
- The Hilbert transform $Hf_m(\ell_k)$ of this density (gives the direction and refinement). It captures the local asymmetry of the spectrum around ℓ_k . If more eigenvalues lie to the left than to the right, Hf_m is negative and ℓ_k is pulled upward by its neighbors, which are larger eigenvalues — more downward correction is therefore needed.

d_k^* is a corrected version of ℓ_k that accounts for its local spectral environment. ANLS adapts the correction to each eigenvalue based on the density and asymmetry of the spectrum at that point.

5 Estimating f_m and Hf_m by KDE: The Epanechnikov Kernel

How can we estimate $f_m(\ell_k)$ and $Hf_m(\ell_k)$ when the true eigenvalue density is unknown? We have access to the p empirical eigenvalues of S . We use KDE (Kernel Density Estimation).

The idea of KDE is to estimate the density at a point ℓ_k by looking at how many eigenvalues are close to ℓ_k , with a weight decreasing with distance.

The authors choose the Epanechnikov kernel, optimal in terms of integrated squared error among all compactly supported kernels:

$$K(u) = \frac{3}{4\sqrt{5}} \left(1 - \frac{u^2}{5}\right) \cdot \mathbf{1}_{\{|u| \leq \sqrt{5}\}}$$

where $u = \frac{\ell_k - \ell_j}{h}$ is the normalized distance between two eigenvalues, and h is the bandwidth.

The kernel equals zero outside $[-\sqrt{5}, \sqrt{5}]$: only eigenvalues within a neighborhood of size $h\sqrt{5}$ around ℓ_k contribute. Inside the kernel, the weight is parabolic, maximal at $u = 0$ and decreasing quadratically toward the edges.

The KDE estimator of the density at ℓ_k is therefore the weighted average of kernels centered on each eigenvalue ℓ_j :

$$\hat{f}_m(\ell_k) = \frac{1}{p \cdot h} \sum_{j=1}^p K\left(\frac{\ell_k - \ell_j}{h}\right).$$

We sum the contributions of all neighboring eigenvalues ℓ_j to ℓ_k .

The factor $\frac{1}{ph}$ normalizes so that the integral of $\hat{f}_m$ equals 1. If many eigenvalues are close to ℓ_k , the estimated density is high and $\hat{f}_m(\ell_k)$ is large.

The asymptotically optimal bandwidth for estimating a density from p points in this context results from a bias-variance trade-off, minimizing the asymptotic integrated squared error according to the authors:

$$h = n^{-1/3}.$$

6 The Hilbert Transform of the Kernel: $HK(u)$

The Hilbert transform $Hf_m(\ell_k)$ is the second ingredient of the formula for d_k^* . It captures the local asymmetry of the spectrum around ℓ_k : the extent to which there are more eigenvalues to the left or to the right of ℓ_k .

The Hilbert transform of a function f is defined by the Cauchy principal value:

$$Hf(\lambda) = \text{p.v.} \int_{-\infty}^{+\infty} \frac{f(t)}{\lambda - t} dt$$

where p.v. denotes the Cauchy principal value, which excludes an infinitesimal neighborhood around $t = \lambda$ and then takes the limit.

The key property is that if f is estimated by KDE with kernel K , then Hf is estimated with the analytic Hilbert transform of that same kernel HK , without any additional numerical computation.

For the Epanechnikov kernel, the Hilbert transform is known in closed form. The logarithmic term captures the asymmetry:

$$HK(u) = \frac{-3u}{10\pi} + \frac{3}{4\sqrt{5}\pi} \left(1 - \frac{u^2}{5}\right) \ln\left(\frac{\sqrt{5} - u}{\sqrt{5} + u}\right) \cdot \mathbf{1}_{\{|u| < \sqrt{5}\}}.$$

The estimator of $Hf_m(\ell_k)$ is therefore:

$$\widehat{Hf}_m(\ell_k) = \frac{1}{p \cdot h} \sum_{j=1}^p HK\left(\frac{\ell_k - \ell_j}{h}\right).$$

Same structure as $\hat{f}_m$, but with HK instead of K . We sum the Hilbert transforms of the kernels centered on each neighboring eigenvalue.

7 Assembly: The Complete Computation of d_k^* from the Data

For each eigenvalue ℓ_k of S , the computation of d_k^* proceeds as follows:

1. Compute the normalized distances. For each pair (k, j) , compute $u_{kj} = \frac{\ell_k - \ell_j}{h}$, the $p \times p$ matrix of normalized distances between all pairs of eigenvalues.
2. Evaluate the kernel and its Hilbert transform. Compute $K(u_{kj})$ and $HK(u_{kj})$ for all pairs.
3. Sum to obtain $\hat{f}_m(\ell_k)$ and $\widehat{Hf}_m(\ell_k)$:

$$\hat{f}_m(\ell_k) = \frac{1}{ph} \sum_{j=1}^p K(u_{kj}), \quad \widehat{Hf}_m(\ell_k) = \frac{1}{ph} \sum_{j=1}^p HK(u_{kj}).$$

4. Compute d_k^* via the ANLS formula:

$$d_k^* = \frac{\ell_k}{\left(\pi c \ell_k \hat{f}_m(\ell_k)\right)^2 + \left(1 - c - \pi c \ell_k \widehat{Hf}_m(\ell_k)\right)^2}.$$

8 Matrix Reconstruction and the Case $p > n$

Once the corrected eigenvalues d_k^* have been computed, we reconstruct the estimator by replacing the eigenvalues of S while keeping its eigenvectors:

$$\hat{\Sigma}_{\text{ANLS}} = \sum_{k=1}^p d_k^* u_k u_k^\top = U \text{diag}(d_1^*, \dots, d_p^*) U^\top.$$

The case $p > n$: singular matrix.

When $p > n$, S has exactly $p - n$ null eigenvalues. These are directions in which there are not enough observations to estimate the variance. They are treated separately.

For null eigenvalues, a common value derived from the harmonic mean of the positive eigenvalues is assigned:

$$d_{\text{null}} = \frac{\bar{\lambda}_{\text{harm}}}{c(c-1)} \quad \text{with} \quad \bar{\lambda}_{\text{harm}} = \frac{p_{\text{nz}}}{\sum_{\ell_k > 0} \frac{1}{\ell_k}}$$

where $\bar{\lambda}_{\text{harm}}$ is the harmonic mean of the nonzero eigenvalues of S . The null eigenvalues receive a common value proportional to this harmonic mean, adjusted by $c(c-1)$.

9 Properties of ANLS

1. **Asymptotic optimality** within the class of rotationally invariant estimators.
2. **Adaptive correction**: each eigenvalue receives its own optimal correction.
3. **Guaranteed positive definiteness**.
4. **Convergence of the estimated density**.

10 Limitations of ANLS

1. **ANLS operates in the covariance space**. The estimator minimizes the squared error $\mathbb{E}[\|\hat{\Sigma} - \Sigma\|_F^2]$, which is the error on the covariance matrix. But in finance, what is directly used is Σ^{-1} , the precision matrix, at the core of Markowitz. Minimizing the error on Σ does not guarantee minimizing the error on Σ^{-1} . A small error on an eigenvalue close to zero produces a massive error on its inverse.
2. **ANLS uses a KDE kernel that introduces a smoothing bias**. Estimating f_m with the Epanechnikov kernel and bandwidth $h = n^{-1/3}$ is an approximation. It is asymptotically exact, but on a finite sample it smooths the spectrum and introduces a boundary bias: extreme eigenvalues are poorly estimated because the kernel spills outside the true support of the spectrum.

Quadratic-Inverse Shrinkage: Ledoit & Wolf (2022)

1 Starting Point: Changing the Optimization Criterion

In finance, the dominant use of Σ goes through its inverse: Markowitz optimization, minimum variance constraints, minimum variance portfolio weights — all these operations involve Σ^{-1} implicitly or explicitly.

The problem is that a small error on an eigenvalue of $\hat{\Sigma}$ can produce a much larger error on the corresponding eigenvalue of $\hat{\Sigma}^{-1}$. If d_k is close to 0, an additive error of ε on d_k gives an error on $\frac{1}{d_k}$ of:

$$\frac{1}{d_k + \varepsilon} \approx \frac{1}{d_k} - \frac{\varepsilon}{d_k^2}.$$

The amplification factor is $\frac{1}{d_k^2}$. Inversion amplifies errors quadratically on small eigenvalues.

Therefore optimizing the error on $\hat{\Sigma}$ guarantees nothing about the error on $\hat{\Sigma}^{-1}$. Since inversion disproportionately amplifies errors on small eigenvalues, QIS proposes to minimize directly the error on the precision matrix:

$$(d_1^*, \dots, d_p^*) = \arg \min_{d_1, \dots, d_p} \mathbb{E} \left[\left\| \left(\sum_{k=1}^p d_k u_k u_k^\top \right)^{-1} - \Sigma^{-1} \right\|_F^2 \right].$$

We seek the eigenvalues d_k of $\hat{\Sigma}$ that yield the best possible estimate of Σ^{-1} . The eigenvectors u_k of S are kept unchanged.

2 Working in the Inverse Space: The Fundamental Change of Variable

QIS performs a fundamental change of variable. Rather than working on the eigenvalues ℓ_k of S , it works on their inverses $\mu_k = \frac{1}{\ell_k}$.

We denote:

$$\mu_k = \frac{1}{\ell_k} \quad (\text{eigenvalues of } S^{-1}) \quad \text{and} \quad \delta_k^* = \frac{1}{d_k^*} \quad (\text{eigenvalues of } \hat{\Sigma}^{-1}).$$

The eigenvectors of S^{-1} are identical to those of S . Only the eigenvalues are inverted. We work directly on the precision.

In the precision space, the large eigenvalues of S^{-1} correspond to the small eigenvalues of S :

the low-variance directions, the most poorly estimated. By working directly on $\mu_k = \frac{1}{\ell_k}$, QIS naturally gives more attention to corrections in these critical directions, without the bias of a posteriori inversion.

The problem thus becomes finding the corrected eigenvalues of $\hat{\Sigma}^{-1}$, denoted $\delta_k^* = \frac{1}{d_k^*}$.

3 The Cauchy Kernel: Why Not Use Epanechnikov?

QIS uses the Cauchy kernel to estimate the density of eigenvalues in the precision space and then compute its Hilbert transform analytically:

$$K_{\text{Cauchy}}(u) = \frac{1}{\pi(1+u^2)}.$$

Unlike the Epanechnikov kernel, which has compact support $[-\sqrt{5}, \sqrt{5}]$, the Cauchy kernel has infinite support and decays slowly to zero. This infinite support is a crucial property in the precision space where eigenvalues can be very dispersed.

The key property of the Cauchy kernel is that its Hilbert transform is exact and analytic with no approximation:

$$HK_{\text{Cauchy}}(u) = \frac{u}{\pi(1+u^2)}.$$

Exact formula, no approximation.

For the Epanechnikov kernel, the Hilbert transform involved a logarithmic term that introduced a numerical bias at the boundary of the support. With Cauchy, the Hilbert transform is simply u times the kernel itself — a perfectly clean relationship.

This gives the Cauchy kernel the following property: the associated analytic signal, i.e., the combination $K + iHK$ in the complex plane, is exactly the Cauchy function itself:

$$K_{\text{Cauchy}}(u) + iHK_{\text{Cauchy}}(u) = \frac{1}{\pi(1+u^2)} + \frac{iu}{\pi(1+u^2)} = \frac{1}{\pi} \cdot \frac{1}{1-iu}.$$

The Cauchy kernel and its Hilbert transform form an exact analytic pair, allowing QIS to compute the Stein shrinker $\hat{\theta}$ and its conjugate $\widehat{H\theta}$ without any numerical approximation.

4 The Stein Shrinker $\hat{\theta}$ and Its Conjugate $\widehat{H\theta}$

In the precision space, QIS defines the smoothed Stein shrinker $\hat{\theta}$. It is the analog of $\hat{f}_m(\ell_k)$ for ANLS but applied to the inverses of the eigenvalues. What was a local concentration (density) in the eigenvalue space in ANLS becomes a relative dispersion (position) in the precision space in QIS.

For each $\mu_i = \frac{1}{\ell_i}$, we look at how the other $\mu_j = \frac{1}{\ell_j}$ distribute around it via the Cauchy kernel:

$$\hat{\theta}_i = \frac{1}{n_2} \sum_{j=1}^{n_2} \frac{\mu_j(\mu_j - \mu_i)}{(\mu_j - \mu_i)^2 + h^2\mu_j^2} \quad \text{with } \mu_j = \frac{1}{\ell_j} \text{ the nonzero empirical precisions.}$$

The denominator weights each neighboring precision μ_j by a Cauchy kernel. The factor μ_j in

the numerator gives more weight to large precisions, which correspond to small eigenvalues — the most sensitive to inversion. It is therefore a measure of the attraction of μ_i toward its neighbors, allowing estimation of the relative position in the precision spectrum.

Its conjugate, the Hilbert transform of the Stein shrinker:

$$\widehat{H\theta}_i = \frac{1}{n_2} \sum_{j=1}^{n_2} \frac{h\mu_j^2}{(\mu_j - \mu_i)^2 + h^2\mu_j^2}.$$

Same structure as $\hat{\theta}$, but with $h\mu_j$ in the numerator instead of the imaginary part of the Cauchy analytic signal. It captures the local amplitude of the precision density around μ_i , and is therefore a measure of the local density of μ_j values around μ_i .

We then define the squared amplitude of the analytic signal:

$$\hat{A}_i^2 = \hat{\theta}_i^2 + \widehat{H\theta}_i^2.$$

This is the squared modulus of the analytic signal associated with the Stein shrinker in the precision space. It combines the position measure of μ_i and the amplitude (density) in the inverse space. It measures the global intensity of noise around μ_i , independently of local asymmetry.

If $\hat{A}_i^2$ is large, either μ_i is extreme, or it lies in a zone of high density, or both. It therefore estimates the shrinkage intensity assigned to the eigenvalue, an adaptive intensity depending on parameters h and c .

5 The Bandwidth h Adapted to Problem Severity

As with ANLS, a bandwidth h must be chosen for the Cauchy kernel. QIS uses an adaptive bandwidth that accounts for both c and p :

$$h = \frac{\left(\min\left(c^2, \frac{1}{c^2}\right) \right)^{0.35}}{p^{0.35}}.$$

Numerator: symmetric around $c = 1$ and decreasing as c moves away from 1 in either direction. It adapts h to the dimensional severity of the problem. The more extreme c is, the smaller h and the finer the smoothing.

Denominator: ensures asymptotic convergence. The exponent 0.35 lies in the interval $(0, 2/5)$ required by the theorem to guarantee consistency of the estimator.

ANLS used $h = n^{-1/3}$, a function of n only. QIS uses a bandwidth that depends jointly on c and p , and therefore adapts to the dimensional regime.

6 The Corrected Eigenvalue Formula: The Quadratic Structure in c

The corrected eigenvalues δ_k^* are given by a formula quadratic in c :

$$\delta_k^* = \underbrace{(1-c)^2 \mu_k}_{\substack{\text{raw precision} \\ \mu_k=1/\ell_k \\ \text{dominates as } c \rightarrow 0}} + \underbrace{2c(1-c) \mu_k \hat{\theta}_k}_{\substack{\text{cross correction} \\ \text{via Stein shrinker} \\ \text{dominates at } c \approx 0.5}} + \underbrace{c^2 \mu_k \hat{A}_k^2}_{\substack{\text{squared amplitude} \\ \text{correction} \\ \text{dominates as } c \rightarrow 1}}.$$

The three coefficients $(1-c)^2$, $2c(1-c)$, c^2 sum to 1 for all c , like a squared trinomial. This is a generalized barycentric combination.

Substituting $\mu_k = \frac{1}{\ell_k}$ and simplifying:

$$d_k^* = \frac{\ell_k}{(1-c)^2 + 2c(1-c) \hat{\theta}_k + c^2 \hat{A}_k^2}.$$

The denominator is a multiplicative correction of ℓ_k . When $c = 0$ (classical regime): $d_k^* = \ell_k$, no correction. When $c > 0$, the denominator is greater than 1 for large eigenvalues (downward correction) and less than 1 for small ones (upward correction).

Where does the quadratic structure come from? The quadratic structure $(1-c)^2$, $2c(1-c)$, c^2 is not arbitrary. It follows directly from the QIS optimization criterion and the Marčenko–Pastur equation applied in the precision space. The quadratic structure is therefore an exact mathematical consequence of optimizing the QIS criterion under the Marčenko–Pastur law.

7 Rescaling to Preserve the Trace

Once the d_k^* have been computed, the authors apply a global rescaling to ensure that the trace of $\hat{\Sigma}_{\text{QIS}}$ equals the trace of S .

This is a consistency constraint: the sum of variances of all assets must remain the same before and after shrinkage.

$$\text{scale} = \frac{\sum_{\ell_k > 0} \ell_k}{\sum_k d_k^*} \implies d_k^{(m)} = \text{scale} \times d_k^*.$$

The numerator is the trace of S restricted to nonzero eigenvalues. The denominator is the sum of the corrected eigenvalues before rescaling. The scale factor globally adjusts all d_k^* so that their sum equals exactly $\text{tr}(S)$.

The formula for d_k^* optimizes the error on Σ^{-1} , not on Σ . Nothing guarantees that $\sum_k d_k^* = \text{tr}(\Sigma)$. Without rescaling, $\hat{\Sigma}_{\text{QIS}}$ could have a trace significantly different from S , which would amount to uncontrolled modification of the global volatility of portfolios. Rescaling corrects this by enforcing consistency with S on the trace.

8 The Case $p > n$: Treatment of Null Eigenvalues

When $p > n$, S has $p - n$ null eigenvalues. These directions carry no information in S . QIS assigns them a common value derived from the arithmetic mean of the nonzero precisions:

$$\delta_0 = \frac{1}{(1 - c)\bar{\nu}} \quad \text{with} \quad \bar{\nu} = \frac{1}{n_2} \sum_{\ell_k > 0} \frac{1}{\ell_k}$$

where $\bar{\nu}$ is the arithmetic mean of the nonzero precisions, and the factor normalizes the contribution per null direction.

9 Final Reconstruction and Properties

We reconstruct $\hat{\Sigma}_{\text{QIS}}$ by replacing the eigenvalues of S with the rescaled $d_k^{(m)}$, while keeping the eigenvectors of S :

$$\hat{\Sigma}_{\text{QIS}} = U \text{diag}(d_1^{(m)}, \dots, d_p^{(m)}) U^\top.$$

Guaranteed properties:

1. **Optimality for inversion:** directly minimizes the error on Σ^{-1} .
2. **Spectral consistency with S :** same eigenvectors.
3. **Guaranteed positive definiteness:** all $d_k^{(m)} > 0$.
4. **No boundary smoothing bias:** the Cauchy kernel with infinite support avoids the boundary bias of the Epanechnikov kernel.

Summary: The Three Estimation Processes

Model 1: Linear Shrinkage — Ledoit & Wolf (2004)

Step 1. Compute the sample covariance and the spherical target. $S = \frac{1}{n}X^\top X$ is the sample covariance. $\bar{s}$ is the average variance and $T = \bar{s}I_p$ is the spherical target:

$$\bar{s} = \frac{1}{p} \text{tr}(S).$$

Step 2. Measure the total dispersion of S around T , signal and noise combined:

$$d^2 = \|S - T\|_F^2 = \underbrace{\sum_i (S_{ii} - \bar{s})^2}_{\text{dispersion of variances around their mean}} + \underbrace{\sum_{i \neq j} S_{ij}^2}_{\text{magnitude of covariances}}.$$

Step 3. Estimate the sampling variance of S , i.e., the share of d^2 due to pure noise:

$$b^2 = \frac{1}{n^2} \left[\sum_{t=1}^n \|x_t\|^4 + n\|S\|_F^2 - 2 \sum_{t=1}^n x_t^\top S x_t \right].$$

Step 4. Compute the optimal shrinkage intensity, the ratio of noise to total dispersion:

$$\rho^* = \frac{b^2}{d^2}.$$

Step 5. Reconstruct the estimator as a convex combination of S and the target T , uniformly compressing all eigenvalues toward $\bar{s}$:

$$\hat{\Sigma}_{\text{LW}} = \rho^* \bar{s} I_p + (1 - \rho^*) S.$$

Model 2: Analytical Nonlinear Shrinkage — Ledoit & Wolf (2020)

Step 1. Spectrally decompose S to extract its eigenvalues and eigenvectors, which are kept for the directions:

$$S = U \operatorname{diag}(\ell_1, \dots, \ell_p) U^\top.$$

Step 2. Set the asymptotically optimal bandwidth for estimating a density from p points:

$$h = n^{-1/3}.$$

Step 3. Estimate the local eigenvalue density around each ℓ_k by KDE with the Epanechnikov kernel, representing the local concentration of the spectrum around ℓ_k :

$$\hat{f}_m(\ell_k) = \frac{1}{p \cdot h} \sum_{j=1}^p K\left(\frac{\ell_k - \ell_j}{h}\right), \quad K(u) = \frac{3}{4\sqrt{5}} \left(1 - \frac{u^2}{5}\right) \cdot \mathbf{1}_{\{|u| \leq \sqrt{5}\}}.$$

Step 4. Estimate the Hilbert transform of this density, representing the local asymmetry of the spectrum around ℓ_k , i.e., the imbalance between eigenvalues to the left and right of ℓ_k :

$$\widehat{Hf}_m(\ell_k) = \frac{1}{p \cdot h} \sum_{j=1}^p HK\left(\frac{\ell_k - \ell_j}{h}\right),$$

$$HK(u) = \frac{-3u}{10\pi} + \frac{3}{4\sqrt{5}\pi} \left(1 - \frac{u^2}{5}\right) \ln\left(\frac{\sqrt{5} - u}{\sqrt{5} + u}\right) \cdot \mathbf{1}_{\{|u| < \sqrt{5}\}}.$$

Step 5. Compute the optimal corrected eigenvalue for each ℓ_k , accounting for its local spectral environment:

$$d_k^* = \frac{\ell_k}{\left(\pi c \ell_k \hat{f}_m(\ell_k)\right)^2 + \left(1 - c - \pi c \ell_k \widehat{Hf}_m(\ell_k)\right)^2}.$$

Step 6. Reconstruct the estimator by replacing the eigenvalues of S with the corrected values, each direction receiving its own correction:

$$\hat{\Sigma}_{\text{ANLS}} = U \operatorname{diag}(d_1^*, \dots, d_p^*) U^\top.$$

Model 3: Quadratic-Inverse Shrinkage — Ledoit & Wolf (2022)

Step 1. Spectrally decompose S and move to the precision space, where large precisions correspond to low-variance directions, the most sensitive to inversion:

$$S = U \operatorname{diag}(\ell_1, \dots, \ell_p) U^\top, \quad \mu_k = \frac{1}{\ell_k} \quad \text{nonzero empirical precisions.}$$

Step 2. Set the adaptive bandwidth, symmetric around $c = 1$ and adapting to the dimensional severity of the problem:

$$h = \frac{\left(\min\left(c^2, \frac{1}{c^2}\right) \right)^{0.35}}{p^{0.35}}.$$

Step 3. Compute the smoothed Stein shrinker in the precision space via the Cauchy kernel, measuring the attraction of μ_i toward its neighbors with more weight on large precisions:

$$\hat{\theta}_i = \frac{1}{n_2} \sum_{j=1}^{n_2} \frac{\mu_j (\mu_j - \mu_i)}{(\mu_j - \mu_i)^2 + h^2 \mu_j^2}.$$

Step 4. Compute the conjugate of the Stein shrinker, its exact analytic Hilbert transform via the Cauchy kernel, representing the local amplitude of the precision density around μ_i :

$$\widehat{H\theta}_i = \frac{1}{n_2} \sum_{j=1}^{n_2} \frac{h \mu_j^2}{(\mu_j - \mu_i)^2 + h^2 \mu_j^2}, \quad \hat{A}_i^2 = \hat{\theta}_i^2 + \widehat{H\theta}_i^2.$$

Step 5. Compute the corrected eigenvalues via the quadratic formula in c , whose three terms correspond to the three dimensional regimes:

$$\delta_k^* = \underbrace{(1-c)^2 \mu_k}_{c \rightarrow 0} + \underbrace{2c(1-c) \mu_k \hat{\theta}_k}_{c \approx 0.5} + \underbrace{c^2 \mu_k \hat{A}_k^2}_{c \rightarrow 1}, \quad d_k^* = \frac{1}{\delta_k^*}.$$

Step 6. Apply a global rescaling so that the trace of the final estimator is consistent with the trace of S :

$$\text{scale} = \frac{\sum_{\ell_k > 0} \ell_k}{\sum_k d_k^*}, \quad d_k^{(m)} = \text{scale} \times d_k^*.$$

Step 7. Reconstruct the estimator by replacing the eigenvalues of S with the rescaled corrected eigenvalues, optimized for inversion:

$$\hat{\Sigma}_{\text{QIS}} = U \operatorname{diag}(d_1^{(m)}, \dots, d_p^{(m)}) U^\top.$$